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FACULTY OF MANAGEMENT STUDIES**

DEPARTMENT OF INFORMATION TECHNOLOGY STUDIES

DEPARTMENT OF MANAGEMENT STUDIES

INTERDISCIPLINARY UNDERGRADUATE PROJECT WORK PROPOSAL

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THIS PROJECT REPORT IS SUBMITTED TO THE DEPARTMENT OF INFORMATION
TECHNOLOGY STUDIES AND DEPARTMENT OF MANAGEMENT STUDIES OF THE
FACULTY OF INFORMATION TECHNOLOGY AND COMMUNICATION STUDIES AND
FACULTY OF MANAGEMENT STUDIES OF THE UNIVERSITY OF PROFESSIONAL
STUDIES, ACCRA, IN PARTIAL FULFILLMENT FOR A BACHELOR OF SCIENCE
DEGREE IN INFORMATION TECHNOLOGY MANAGEMENT BACHELOR OF SCIENCE
IN BUSINESS ADMINISTRATION

TOPIC

**DESIGN AND DEVELOPMENT OF AN AI-POWERED WEB APPLICATION FOR
MONITORING ENERGY EXPENDITURE AND DAILY CALORIC NEEDS**

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ABSTRACT

The increasing adoption of digital health technologies has created a growing need for accurate, personalized, and accessible fitness tracking tools. Many existing applications rely on generalized algorithms that fail to account for individual metabolic variations, resulting in inaccurate calorie estimations and limited guidance for users seeking to manage their weight. This study proposes the design and development of an AI-powered web application that estimates calories burned during workouts and calculates daily caloric needs for weight loss, maintenance, or weight gain. The system incorporates standardized metabolic equivalent (MET) values and the Mifflin-St. Jeor equation to enhance accuracy, while hybrid AI logic provides personalized recommendations based on user attributes and activity levels. A Design and Development Research (DDR) approach, supported by agile methodology, guided the creation and iterative refinement of the system. Data collection involved usability testing, questionnaires, and observational analysis among students of the University of Professional Studies, Accra. The system was developed using HTML, CSS, JavaScript, and Firebase for cloud-based storage and authentication. Findings indicate that integrating AI-driven logic with established metabolic

formulas improves the accuracy and relevance of caloric insights compared to generic fitness applications. The study contributes to digital health innovation by demonstrating how personalized, web-based solutions can enhance user engagement and support informed fitness and nutritional decisions. The proposed system also provides practical value for individuals, fitness professionals, and health-focused enterprises seeking data-driven approaches to wellness management.

INTRODUCTION

1.1 BACKGROUND OF THE STUDY

In recent years, the integration of information technology into healthcare and fitness management has profoundly transformed the way individuals monitor, evaluate, and improve their physical well-being. Rapid technological advancements, combined with increased internet accessibility, have enabled the development of digital health solutions that support individuals in managing their health more effectively. At the same time, modern lifestyles characterized by reduced physical activity, prolonged screen time, and unhealthy dietary habits have contributed to a global rise in lifestyle-related health conditions. According to the World Health Organization (WHO, 2022), physical inactivity and poor nutrition are among the leading risk factors for non-communicable diseases such as obesity, type 2 diabetes, and cardiovascular diseases. These trends have intensified global interest in personal health management tools that empower individuals to take greater control of their fitness and nutritional behaviors.

One of the most critical concepts underlying effective health and fitness management is energy balance, which refers to the relationship between calories consumed through food and beverages and calories expended through basal metabolism and physical activity. Energy balance plays a central role in weight management, athletic performance, and overall metabolic health. When caloric intake exceeds energy expenditure, weight gain is likely to occur, whereas sustained caloric deficits can lead to weight loss. Hall et al. (2012) emphasize that understanding and managing energy balance is fundamental to achieving long-term health and fitness goals. Consequently, accurate monitoring of both caloric intake and energy expenditure is essential for individuals seeking to improve or maintain their physical well-being.

Energy expenditure is influenced by multiple factors, including age, sex, body composition, height, weight, and the intensity and duration of physical activity. Daily caloric needs vary significantly from one individual to another, making generalized recommendations insufficient for many users. Traditional methods of estimating caloric expenditure, such as static equations or generic activity charts, often fail to capture these individual differences accurately. As a result, users may receive misleading feedback that undermines the effectiveness of fitness programs and dietary planning. This challenge highlights the importance of developing systems capable of delivering personalized and context-aware calorie estimations.

The increasing adoption of digital health technologies has led to the widespread use of calorie tracking and fitness monitoring applications. These tools are designed to help users track physical activity, estimate energy expenditure, and monitor dietary intake in a convenient and accessible manner. Wearable devices, mobile applications, and web-based platforms now play a central role in modern health management practices. While these technologies have significantly

improved accessibility and user engagement, many existing solutions still suffer from notable limitations. Research suggests that a large proportion of fitness applications rely on simplified formulas, limited datasets, and non-adaptive algorithms, which can compromise accuracy and personalization (Krebs & Duncan, 2015).

Advancements in artificial intelligence (AI) and machine learning have introduced new possibilities for overcoming these limitations. AI-powered systems can process large volumes of data and identify complex patterns that traditional computational approaches may overlook. In the context of health and fitness management, AI can analyze user-specific parameters such as age, weight, height, activity type, exercise intensity, and frequency to generate more accurate and individualized estimates of energy expenditure and daily caloric requirements. Chen and Zhang (2020) note that intelligent health systems leveraging AI are better equipped to deliver personalized recommendations, adapt to user behavior over time, and support informed decision-making.

Web-based platforms further enhance the practicality of AI-powered health applications by providing cross-device accessibility and ease of use. Unlike mobile-only applications or systems dependent on specialized hardware, web applications can be accessed through standard browsers on desktops, tablets, and smartphones. This flexibility reduces barriers to adoption and promotes consistent usage. Additionally, modern web technologies support real-time data processing, interactive user interfaces, and secure data management, all of which are essential features for effective digital health solutions (Mozilla Developer Network, 2023).

Despite these technological advancements, several challenges remain in the design and implementation of effective calorie monitoring systems. One major concern is transparency in

computational logic. Many users are unaware of how calorie estimates are generated, leading to skepticism and reduced trust in the system's outputs. Opaque algorithms and black-box models can discourage long-term engagement, particularly when users observe discrepancies between estimated and actual outcomes. Furthermore, privacy and data security concerns remain critical issues, as health-related data is highly sensitive and requires responsible handling.

Another limitation observed in existing systems is their lack of adaptability to individual lifestyle changes. Many applications fail to update caloric recommendations dynamically as users modify their activity levels, fitness goals, or physical characteristics. This static approach reduces the relevance and effectiveness of the feedback provided. Consequently, there is a growing demand for intelligent systems that combine scientifically validated metabolic models with adaptive AI techniques and transparent logic to deliver accurate, user-centered solutions.

In response to these challenges, the development of an AI-powered web application for monitoring energy expenditure and daily caloric needs presents a promising approach. By integrating established metabolic equations with intelligent data processing techniques, such a system can provide reliable and personalized insights into users' fitness and nutritional requirements. The use of AI enables continuous refinement of recommendations based on user input and behavior, while a web-based architecture ensures accessibility, scalability, and ease of use.

This study therefore focuses on the design and development of an AI-powered web application aimed at improving the accuracy, usability, and transparency of energy expenditure and daily caloric need estimation. The proposed system seeks to support individuals engaged in weight management and physical fitness by providing data-driven decision support and personalized

feedback. By addressing the limitations of existing solutions and leveraging modern web and AI technologies, this research contributes to the ongoing efforts to enhance digital health tools and promote healthier lifestyles in an increasingly technology-driven society.

1.2 STATEMENT OF THE PROBLEM

Accurate estimation of energy expenditure and daily caloric requirements remains a persistent challenge for individuals seeking to maintain, lose, or gain weight effectively. Energy expenditure is a complex physiological process influenced by numerous interrelated factors, including basal metabolic rate, body composition, age, sex, genetic predisposition, and the type, intensity, and duration of physical activity. Despite the growing availability of digital fitness and calorie tracking applications, many existing systems fail to capture this complexity adequately. Instead, they rely heavily on generalized equations and simplified assumptions that overlook individual variability, leading to inaccurate estimations and suboptimal guidance (Li & Mensah, 2023).

Most commercially available calorie tracking systems utilize standardized metabolic formulas or average values derived from population-level data. While these approaches may offer rough estimates suitable for general use, they often lack the precision required for personalized health and fitness management. For example, individuals with higher muscle mass typically have greater resting energy expenditure than those with similar body weight but higher fat composition. Similarly, variations in metabolic efficiency and exercise intensity can significantly affect caloric burn, yet these differences are rarely accounted for in basic tracking tools. As a

result, users may receive misleading feedback that does not reflect their true physiological state, potentially undermining weight management efforts and overall health outcomes.

Inaccurate calorie estimation can have serious implications for users' decision-making processes. Individuals attempting to lose weight may unknowingly consume excess calories due to overestimated energy expenditure, while those aiming to gain muscle mass may fail to achieve sufficient caloric intake because of underestimated requirements. Over time, such discrepancies can lead to frustration, reduced motivation, and abandonment of health-related goals. Li and Mensah (2023) highlight that inaccurate feedback from digital health tools can negatively influence user confidence and adherence, ultimately limiting the long-term effectiveness of these technologies.

Another significant limitation of many existing fitness and calorie tracking applications is their narrow focus on exercise-related calorie expenditure without adequately integrating daily caloric needs. While tracking calories burned during physical activity is valuable, it represents only one component of total daily energy expenditure. Basal metabolic processes, thermic effect of food, and non-exercise physical activities collectively account for a substantial proportion of daily caloric usage. However, many systems fail to present users with a holistic view of their energy balance, leaving them without clear guidance on how caloric intake should be adjusted in relation to overall expenditure.

Consequently, users are often left uncertain about whether they should maintain a calorie deficit, surplus, or balance to achieve specific fitness objectives such as weight loss, weight maintenance, or muscle gain. Without clear and actionable insights, individuals may struggle to translate raw numerical data into meaningful lifestyle changes. Ainsworth et al. (2011) argue that effective

health management tools must go beyond simple data reporting and provide interpretive guidance that supports informed decision-making and sustainable behavior change. The absence of such guidance significantly reduces the practical usefulness of many current systems.

In addition to limited scope and personalization, transparency remains a critical issue in the design of calorie estimation technologies. Many modern fitness applications employ proprietary algorithms or complex computational models that operate as “black boxes,” offering little to no explanation of how calorie values are generated. While these systems may leverage sophisticated techniques, the lack of interpretability can erode user trust, particularly when estimated outcomes do not align with personal experiences or expectations. Gao et al. (2015) emphasize that transparency in health-related systems is essential for building user confidence and encouraging long-term engagement.

When users do not understand how recommendations are derived, they are less likely to rely on or act upon them consistently. This issue is especially problematic in health and fitness contexts, where sustained behavioral change is required to achieve meaningful results. Transparent systems that clearly communicate underlying assumptions, data inputs, and calculation methods can empower users to better understand their bodies and make more informed choices. Unfortunately, such transparency is often sacrificed in favor of simplicity or proprietary advantage in many commercial applications.

Accessibility also presents a notable challenge in the current landscape of calorie tracking and fitness monitoring tools. A significant number of systems depend on mobile-only platforms or specialized wearable devices to collect activity data. While these technologies offer convenience and real-time monitoring, they may not be universally accessible due to cost, technical

limitations, or user preference. Individuals in resource-limited settings or those who prefer browser-based solutions may be excluded from benefiting fully from such systems. This reliance on specific hardware or operating systems restricts inclusivity and limits the broader adoption of digital health technologies.

Moreover, wearable-dependent systems often require continuous device usage and maintenance, such as regular charging and calibration, which can negatively affect user compliance. When data collection becomes burdensome, users are more likely to discontinue use, reducing the long-term effectiveness of the application. A web-based solution that minimizes dependency on specialized hardware while still delivering accurate and personalized insights can address these barriers and enhance accessibility for a wider range of users.

Given these challenges, there is a clear and pressing need for a more comprehensive approach to energy expenditure and caloric need estimation. An ideal system should integrate scientifically validated metabolic equations with intelligent, adaptive logic capable of accounting for individual physiological characteristics and lifestyle factors. The incorporation of artificial intelligence offers significant potential in this regard, as AI techniques can analyze user-specific data, identify patterns, and refine estimations over time based on observed behaviors and outcomes.

Equally important is the need for transparency and user-centered design. A system that clearly explains how calorie estimates are generated and how recommendations align with user goals can foster trust and encourage sustained engagement. By presenting information in an accessible and interpretable manner, such a system can bridge the gap between complex physiological concepts and everyday health decisions.

Therefore, the development of a web-based, AI-powered system for monitoring energy expenditure and daily caloric needs represents a promising solution to the limitations of existing tools. By combining accessibility, personalization, transparency, and scientific validity, such a system can provide accurate, actionable, and meaningful support for individuals pursuing weight management and physical fitness goals. Addressing these gaps not only enhances the practical utility of digital health technologies but also contributes to broader efforts aimed at promoting healthier lifestyles and improving long-term health outcomes.

1.3 RESEARCH OBJECTIVES

General Objective: The general objective of this study is to design and develop an AI-powered web application that accurately monitors energy expenditure and estimates daily caloric needs to support effective weight management and informed fitness decision-making.

Specific Objectives:

IT-Related Objectives

1. To design a responsive web interface using HTML, CSS, and JavaScript that allows users to log workouts and track caloric data (ISO 9241-11).
2. To implement hybrid AI techniques using MET values and metabolic formulas for accurate calorie expenditure estimation (Ainsworth et al., 2011; Mifflin et al., 1990).

Non-IT Objectives

1. To help users make informed health and nutrition decisions through accurate, personalized caloric guidance (WHO, 2020).
2. To promote consistent exercise habits by providing clear performance feedback and motivational insights (Krebs & Duncan, 2015).

Research Questions:

This study is guided by the following research questions:

1. How can a web-based system accurately estimate energy expenditure during physical activity?
2. How can daily caloric needs be determined to support weight maintenance, weight loss, or weight gain?
3. To what extent does an AI-powered web application improve usability and user satisfaction compared to existing calorie tracking systems?

Based on these research questions, the study is guided by the following hypothesis:

- H_1 : An AI-powered web application significantly improves the accuracy and usability of calorie monitoring compared to conventional calorie tracking systems.

1.4 PURPOSE OF THE STUDY

The purpose of this study is to design and develop a scientifically grounded and technologically advanced web-based application capable of accurately estimating calories burned during physical activity and determining users' daily caloric requirements in alignment with their individual health and fitness goals. The proposed system seeks to address persistent limitations in existing digital fitness platforms, particularly those related to estimation accuracy, personalization, transparency, and accessibility. By integrating established metabolic principles with modern web and artificial intelligence technologies, the study aims to provide a reliable and user-centered solution for energy expenditure monitoring and calorie management.

Accurate calculation of calories burned during workouts is essential for effective weight management and fitness planning. Energy expenditure varies significantly based on factors such as age, sex, body weight, height, exercise type, intensity, and duration. Many existing applications rely on generalized formulas or average values that do not adequately reflect individual physiological differences, leading to misleading feedback and reduced effectiveness. This study aims to overcome these shortcomings by implementing scientifically validated metabolic equations and intelligent computational logic that account for user-specific attributes and activity parameters. Through this approach, the application is expected to deliver more precise and personalized estimations of workout-related energy expenditure.

In addition to exercise-based calorie tracking, the study emphasizes the importance of determining daily caloric requirements as part of a comprehensive energy balance framework. Daily caloric needs are influenced not only by physical activity but also by basal metabolic rate, lifestyle factors, and personal fitness objectives such as weight loss, weight maintenance, or muscle gain. The proposed system aims to integrate these factors to provide users with clear and actionable guidance on caloric intake. By aligning caloric recommendations with individual goals, the application seeks to support informed decision-making and promote sustainable health behaviors.

Personalization is a central objective of the study. Rather than offering static or generic outputs, the application is designed to adapt recommendations based on user-provided data and goal selection. This adaptive approach enhances relevance and usability, enabling users to better understand how their daily choices impact energy balance. Furthermore, the system prioritizes transparency by clearly presenting the underlying assumptions, inputs, and logic used in calorie calculations, thereby fostering user trust and engagement.

Accessibility is another key focus of the study. By adopting a web-based architecture, the application eliminates dependence on mobile-only platforms or specialized wearable devices, making it accessible across a wide range of devices and environments. This design choice supports broader adoption and ensures that users can conveniently access the system regardless of technological constraints.

Overall, the study aims to contribute to the advancement of digital health and fitness management by developing an intelligent, accessible, and scientifically valid web application. By addressing critical gaps in existing platforms, the proposed system seeks to enhance the accuracy

of energy expenditure estimation, improve user understanding of caloric needs, and support effective weight management and physical fitness outcomes.

1.5 SIGNIFICANCE OF THE STUDY

The significance of this study can be examined from multiple interrelated perspectives, including technological, practical, academic, and societal dimensions. By addressing critical limitations in existing digital fitness and calorie monitoring systems, the study offers meaningful contributions to the advancement of digital health technologies and their role in promoting healthier lifestyles. The proposed AI-powered web application not only enhances the accuracy and usability of calorie estimation but also supports informed decision-making and long-term behavioral change.

From an information technology perspective, this study demonstrates the effective application of artificial intelligence and modern web technologies within the domain of digital health systems. The integration of AI-driven logic with scientifically validated metabolic equations highlights how intelligent systems can move beyond static, rule-based calculations toward more adaptive, data-driven solutions. Unlike conventional fitness applications that rely on fixed formulas, the proposed system leverages user-specific parameters to generate personalized outputs, thereby illustrating the practical value of hybrid AI approaches in health-related decision support systems (Chen & Zhang, 2020).

Additionally, the study contributes to the growing emphasis on transparency and interpretability in AI-based systems. In health applications, trust is a critical factor influencing user adoption and continued use. By incorporating clear and explainable computational logic, the system addresses concerns associated with black-box models and opaque algorithms. This approach aligns with current best practices in responsible AI development, where explainability, reliability, and user understanding are prioritized. From a technological standpoint, the study also demonstrates how web-based architectures can be effectively used to deliver scalable, cross-platform health solutions without reliance on specialized hardware or proprietary mobile ecosystems.

From a practical perspective, the significance of this study lies in its potential to directly support individuals in managing their physical activity, nutrition, and overall health more effectively. Accurate and personalized information is essential for users seeking to achieve fitness goals such as weight loss, weight maintenance, or muscle gain. Many existing applications provide fragmented or misleading feedback, which can confuse users and hinder progress. The proposed

system addresses this issue by offering integrated insights into calories burned during workouts and daily caloric requirements, presented in a clear and actionable manner.

By explicitly indicating whether a user is operating within a calorie deficit, surplus, or balance, the application enables users to understand the practical implications of their lifestyle choices. This clarity empowers individuals to make informed adjustments to their exercise routines and dietary habits. As emphasized by Hall et al. (2012), effective weight management depends not only on tracking data but also on understanding energy balance and applying that knowledge consistently. The system's ability to translate complex metabolic concepts into user-friendly feedback enhances its practical usefulness and supports sustainable behavior change.

Furthermore, the web-based nature of the application improves accessibility and convenience for a wide range of users. Unlike mobile-only or wearable-dependent solutions, the proposed system can be accessed through standard web browsers on various devices, reducing barriers related to cost, device compatibility, and technical expertise. This accessibility increases the likelihood of consistent usage, which is essential for achieving meaningful health outcomes. From a practical standpoint, the study therefore contributes to the development of inclusive digital health tools that cater to diverse user populations.

Academically, this study contributes to the expanding body of literature on digital health technologies, fitness monitoring systems, and the application of artificial intelligence in healthcare contexts. It provides a structured framework for integrating metabolic science with AI-driven computation and web-based delivery, which can inform future research and system development. By addressing gaps in personalization, transparency, and accessibility, the study

advances scholarly understanding of how intelligent health systems can be designed to better meet user needs.

The research also contributes to discussions on technology adoption and user engagement in digital health platforms. Previous studies have shown that perceived usefulness, trust, and ease of use significantly influence the adoption and sustained use of health technologies. By prioritizing accuracy, explainability, and user-centered design, the proposed system offers empirical and conceptual insights that can guide future investigations into user behavior and system effectiveness. Additionally, the study can serve as a reference point for researchers exploring similar applications in related domains, such as chronic disease management, nutritional planning, and preventive healthcare.

From a societal perspective, the significance of this study is closely tied to its potential impact on public health and preventive healthcare strategies. Lifestyle-related health conditions, including obesity, cardiovascular diseases, and metabolic disorders, continue to pose significant challenges to healthcare systems worldwide. The World Health Organization (WHO, 2022) emphasizes the importance of preventive measures and individual awareness in reducing the burden of non-communicable diseases. By promoting understanding of energy balance and encouraging healthier lifestyle choices, the proposed system supports these broader public health objectives.

The application fosters self-awareness and personal responsibility by enabling users to monitor and interpret their own health-related data. Over time, increased awareness of caloric needs and energy expenditure can lead to healthier habits, reduced risk of chronic diseases, and improved overall well-being. On a larger scale, widespread adoption of accessible and reliable digital

health tools can contribute to reduced healthcare costs by minimizing the need for medical interventions associated with preventable conditions.

Moreover, the societal value of the study extends to its emphasis on equitable access to health information. By offering a web-based solution that does not depend on expensive devices or advanced technical infrastructure, the system supports inclusivity and digital equity. This approach aligns with global efforts to leverage technology as a means of improving health outcomes across diverse populations.

In summary, the significance of this study lies in its multifaceted contributions across technological, practical, academic, and societal domains. Technologically, it demonstrates the effective integration of AI and web technologies in transparent and reliable health systems. Practically, it empowers users with accurate, personalized, and actionable insights for managing fitness and nutrition. Academically, it advances research on intelligent digital health applications and provides a foundation for future studies. Societally, it supports preventive healthcare and promotes healthier lifestyles, contributing to long-term improvements in public health outcomes.

1.6 CHAPTER OUTLINE

This documentation is systematically structured into three main chapters, each designed to address specific aspects of the study in a logical and coherent manner. The organization of the chapters ensures clarity, continuity, and a comprehensive presentation of the research process, from conceptual foundations to methodological implementation.

Chapter One introduces the study by providing a detailed background that establishes the context and motivation for the research. It outlines the growing importance of digital health technologies

and highlights the challenges associated with accurate calorie estimation and fitness monitoring. This chapter clearly defines the problem statement, identifying gaps in existing digital fitness platforms that necessitate the development of a more accurate, personalized, and accessible solution. In addition, the chapter presents the objectives of the study, which articulate the intended outcomes and guide the overall research direction. The research questions are also outlined to frame the inquiry and ensure alignment between the identified problem and the proposed solution. Furthermore, Chapter One discusses the significance of the study from technological, practical, academic, and societal perspectives, emphasizing its contributions to digital health innovation and preventive healthcare. The chapter concludes with an overview of the overall structure of the documentation, providing readers with a clear roadmap of the content presented in subsequent chapters.

Chapter Two focuses on a comprehensive review of relevant theoretical and empirical literature related to the study. This chapter examines existing calorie tracking and fitness monitoring systems, highlighting their strengths, limitations, and areas requiring improvement. It explores key theoretical concepts such as energy balance, metabolic rate, and models used for estimating energy expenditure and daily caloric needs. Additionally, the chapter reviews the application of artificial intelligence in health-related systems, with particular emphasis on personalization, decision support, and transparency in AI-driven applications. Technology adoption models are also discussed to provide insights into factors influencing user acceptance and sustained use of digital health tools. By synthesizing findings from previous studies, Chapter Two establishes a strong theoretical foundation for the research and identifies gaps that the current study seeks to address.

Chapter Three describes the methodology adopted for the study, detailing the systematic approach used to achieve the research objectives. This chapter outlines the research design and research approach, explaining how they align with the study's goals. It describes the study population and sampling techniques employed to ensure representative and reliable data collection. The chapter further discusses the data collection methods used, including tools, procedures, and ethical considerations. Finally, the methods of data analysis are presented, explaining how collected data are processed, analyzed, and interpreted to address the research questions. Together, these methodological components ensure the validity, reliability, and rigor of the study.

Overall, the structured organization of this documentation supports a clear and comprehensive presentation of the research, enabling readers to understand the study's purpose, theoretical grounding, and methodological execution in a logical and cohesive manner.

1.7 CONCLUSION

This chapter has provided a comprehensive introduction to the study and established a strong foundation for the development of an AI-powered web application aimed at monitoring energy expenditure and determining daily caloric needs. The background of the study highlighted the increasing integration of information technology into health and fitness management, as well as the growing global concern over lifestyle-related health conditions. By situating the research within this broader context, the chapter emphasized the relevance and timeliness of developing intelligent digital solutions that support accurate, personalized, and accessible health monitoring.

The chapter clearly identified the challenges associated with existing calorie tracking and fitness monitoring systems. Key limitations such as generalized calorie estimation methods, lack of personalization, limited transparency, and restricted accessibility were discussed in detail. These challenges were shown to negatively affect the accuracy, usability, and trustworthiness of current platforms, ultimately reducing their effectiveness in supporting long-term health and fitness goals. By outlining these issues, the chapter justified the need for a more advanced and user-centered approach that integrates artificial intelligence with scientifically validated metabolic models.

Furthermore, the objectives of the study were clearly articulated, providing a focused direction for the research and system development process. These objectives emphasized the design of an accurate, transparent, and accessible web-based application capable of delivering personalized insights into energy expenditure and daily caloric requirements. The research questions were also presented to guide the investigation and ensure alignment between the identified problem, the proposed solution, and the expected outcomes of the study. Together, the objectives and research

questions established a clear framework for evaluating the effectiveness and contribution of the proposed system.

The significance of the study was examined from multiple perspectives, including information technology, practical application, academic contribution, and societal impact. From a technological standpoint, the chapter highlighted the application of artificial intelligence and modern web technologies in digital health systems. Practically, the study was shown to support individuals in making informed decisions regarding physical activity and nutrition. Academically, the research contributes to the growing body of literature on AI-powered health applications and digital fitness management. From a societal perspective, the study aligns with preventive healthcare goals by promoting awareness of energy balance and encouraging healthier lifestyle choices.

In conclusion, this chapter has successfully established the rationale, scope, and importance of the study, laying the groundwork for subsequent chapters. By clearly defining the problem, objectives, and significance, it provides a coherent starting point for the research. The next chapter presents a comprehensive review of related theoretical and empirical literature, which serves as the foundation for the methodological design and system development discussed in later chapters.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

This chapter presents a comprehensive review of existing scholarly literature relevant to the design and development of an AI-powered web application for monitoring energy expenditure and daily caloric needs. The review is intended to situate the study within the broader context of digital health and fitness management research, while providing the theoretical and empirical foundations necessary to inform system design and methodological choices. By critically examining prior studies, models, and technologies, this chapter establishes a clear academic basis for the proposed solution and identifies gaps that the current study seeks to address.

The literature review is structured around three major thematic areas. The first area focuses on the theoretical concepts underlying calorie estimation, energy balance, and the application of artificial intelligence in digital health systems. This section explores foundational physiological principles such as basal metabolic rate, total daily energy expenditure, and factors influencing caloric burn during physical activity. It also examines commonly used metabolic equations and computational models for estimating energy expenditure and daily caloric needs. In addition, the role of artificial intelligence in enhancing personalization, adaptability, and decision support in health-related applications is discussed, highlighting how AI techniques can improve accuracy and user relevance when compared to traditional rule-based approaches.

The second area of the literature review examines empirical studies related to fitness monitoring and calorie tracking technologies. This section analyzes existing research on mobile applications, wearable devices, and web-based platforms used for tracking physical activity, caloric intake, and energy expenditure. Emphasis is placed on evaluating the effectiveness, accuracy, and usability of these systems, as well as identifying common limitations such as reliance on generalized estimations, lack of transparency, and restricted accessibility. Findings from empirical studies are synthesized to highlight trends, challenges, and best practices in the development of digital fitness tools, thereby providing practical insights that inform the design of the proposed system.

The third area of the review focuses on theoretical frameworks related to technology adoption and user engagement. This section discusses relevant models such as the Technology Acceptance Model (TAM), Unified Theory of Acceptance and Use of Technology (UTAUT), and other user-centered design theories that explain factors influencing the adoption and sustained use of digital health applications. Concepts such as perceived usefulness, ease of use, trust, and transparency are examined in relation to fitness and health monitoring systems. Understanding these factors is critical for designing an application that not only delivers accurate information but also encourages consistent user engagement and long-term behavior change.

Overall, the purpose of this chapter is to establish a strong academic foundation for the study by synthesizing relevant theoretical and empirical literature. By identifying gaps in existing research and limitations in current calorie tracking systems, the chapter justifies the need for an AI-powered, web-based solution that emphasizes accuracy, personalization, transparency, and accessibility. The insights gained from this review directly inform the design decisions and

methodological approach presented in subsequent chapters, ensuring that the proposed system is grounded in established research while addressing unmet needs in the field of digital health and fitness management.

2.2 Theoretical Literature

Theoretical literature provides the conceptual foundation for understanding how energy expenditure is calculated, how daily caloric needs are estimated, and how intelligent systems can effectively support decision-making in health and fitness applications. These theoretical constructs are essential for the design of accurate, reliable, and user-centered digital fitness systems. By grounding application development in established physiological models and intelligent system theories, developers can ensure that outputs are both scientifically valid and practically meaningful.

A central concept in the estimation of energy expenditure during physical activity is the Metabolic Equivalent of Task (MET). MET is defined as the ratio of the rate of energy expended during a specific activity to the rate of energy expended at rest. One MET is approximately equal to the energy cost of sitting quietly, typically quantified as 1 kcal/kg/hour. According to Ainsworth et al. (2011), MET values provide a standardized and empirically validated approach for quantifying the intensity of a wide range of physical activities, from light activities such as walking to vigorous exercises such as running or competitive sports.

The use of MET values has become widespread in exercise science, epidemiological research, and digital health applications due to several advantages. First, MET-based estimation offers simplicity and flexibility, allowing energy expenditure to be calculated using readily available variables such as body weight, activity duration, and activity intensity. Second, MET values are derived from extensive experimental data and are cataloged in comprehensive compendiums, making them a reliable reference for developers of fitness applications. By incorporating MET values into calorie tracking systems, physical activities can be translated into meaningful caloric expenditure estimates that users can easily interpret and apply in their daily routines.

Despite their utility, MET-based estimations must be contextualized within individual physiological characteristics. Factors such as fitness level, movement efficiency, and environmental conditions can influence actual energy expenditure. Nevertheless, MET remains a widely accepted theoretical model because it balances scientific rigor with practical applicability. In digital health systems, MET values serve as a foundational layer upon which more personalized and adaptive logic can be built.

In addition to activity-based energy expenditure, estimating daily caloric needs requires a thorough understanding of Basal Metabolic Rate (BMR). BMR represents the minimum amount of energy required to sustain essential physiological processes at rest, including respiration, circulation, cellular metabolism, and thermoregulation. For most individuals, BMR accounts for the largest proportion of total daily energy expenditure. Accurate estimation of BMR is therefore critical for determining overall caloric requirements and achieving effective energy balance management.

Several predictive equations have been developed to estimate BMR, with the Harris-Benedict and Mifflin-St Jeor equations being among the most widely used. While the Harris-Benedict equation has historical significance, subsequent research has shown that it may overestimate BMR in certain populations. In contrast, the Mifflin-St Jeor equation has been demonstrated to provide more accurate and reliable estimates for healthy adults and is widely recommended in both clinical nutrition and fitness contexts (Mifflin et al., 1990). The equation incorporates key individual variables such as age, sex, weight, and height, allowing for more personalized estimations than generalized averages.

Integrating BMR with physical activity data enables the calculation of Total Daily Energy Expenditure (TDEE), which represents the total number of calories an individual burns in a day. TDEE forms the basis for determining whether an individual is operating in a caloric deficit, surplus, or balance. This information is essential for supporting specific fitness goals such as weight loss, weight maintenance, or muscle gain. From a theoretical standpoint, the combination of BMR and activity-related energy expenditure reflects the core principle of energy balance, which underpins modern weight management and metabolic health strategies.

Beyond physiological models, artificial intelligence plays an increasingly important role in enhancing the functionality and effectiveness of digital health applications. AI-based systems are capable of processing user data, identifying patterns, and generating personalized recommendations that adapt to individual needs and goals. In the context of fitness and calorie tracking, AI enables systems to move beyond static calculations and provide dynamic feedback that reflects user behavior over time.

However, the application of AI in health-related systems requires careful consideration of transparency, interpretability, and trust. Many advanced machine learning models operate as black boxes, producing outputs without clear explanations of how decisions are made. While such models may offer high predictive accuracy, their lack of interpretability can be problematic in health contexts where users must understand and trust system recommendations. Chen and Zhang (2020) argue that hybrid AI systems, which combine rule-based logic with deterministic, formula-driven computation, are particularly well suited for health applications.

Hybrid AI approaches allow systems to maintain scientific validity by grounding calculations in established physiological equations while also incorporating intelligent logic for personalization and decision support. This combination enhances transparency by enabling both users and developers to trace how outputs are generated. In fitness applications, such transparency improves user confidence, encourages consistent engagement, and supports informed decision-making. From a theoretical perspective, hybrid AI systems represent a balanced approach that aligns technological innovation with ethical and usability considerations.

From a technological standpoint, the theoretical foundations of modern web-based health applications are closely linked to advancements in frontend and backend development technologies. Frontend technologies such as HTML, CSS, and JavaScript provide the structural, visual, and interactive components of web applications. These technologies enable the development of responsive and user-friendly interfaces that can adapt to different devices and screen sizes, thereby improving accessibility and usability.

On the backend, cloud-based platforms such as Firebase offer robust solutions for data storage, authentication, and real-time synchronization. Firebase supports secure user authentication,

scalable databases, and seamless integration with web applications, making it particularly suitable for health-related systems that handle sensitive user data. According to Mozilla Developer Network (2023) and Firebase (2024), such cloud-based architectures support high availability, data consistency, and efficient system performance.

The integration of modern web technologies with AI-driven logic creates a theoretical and technical framework that supports accessibility, scalability, and reliability. Web-based architectures eliminate dependence on mobile-only platforms or specialized hardware, enabling users to access health applications across a wide range of devices and environments. This technological foundation aligns with broader digital health goals of inclusivity and widespread adoption.

In summary, the theoretical literature reviewed in this section provides a comprehensive foundation for the design and development of an AI-powered web application for monitoring energy expenditure and daily caloric needs. Physiological models such as MET and BMR offer scientifically validated methods for estimating energy expenditure and caloric requirements, while hybrid AI approaches enhance personalization and transparency. Modern web technologies further support accessibility, scalability, and usability. Together, these theoretical perspectives inform the development of a robust, user-centered digital health system capable of supporting effective fitness and health management.

2.3 Empirical Literature

Empirical studies have extensively examined the effectiveness, usability, and adoption of calorie tracking and fitness monitoring systems, providing valuable insights into how digital health technologies influence user behavior and health outcomes. As the use of mobile and web-based fitness applications continues to grow, researchers have increasingly focused on evaluating the accuracy, personalization, user engagement, and long-term sustainability of these systems. The findings from empirical research highlight both the potential benefits of digital fitness tools and the persistent challenges that limit their effectiveness.

Krebs and Duncan (2015) conducted a large-scale national survey examining the adoption and usage patterns of mobile health applications in the United States. Their findings revealed that while fitness and calorie tracking applications are among the most widely downloaded categories of health apps, long-term user engagement remains relatively low. A significant proportion of users reported discontinuing app usage after a short period. The study identified several contributing factors to this decline in engagement, including inaccurate calorie estimates, limited personalization, and insufficient feedback mechanisms. Users expressed frustration when application outputs did not align with their lived experiences or fitness outcomes. These findings underscore the importance of accuracy and personalized insights as critical determinants of sustained usage and user satisfaction.

Building on these observations, Li and Mensah (2023) conducted a comparative analysis of several popular fitness tracking systems to assess the accuracy of calorie expenditure estimates. Their study compared application-generated calorie estimates with measurements obtained through standardized physiological assessment methods. The results revealed significant discrepancies between estimated and actual calorie expenditure across multiple applications.

Systems that relied solely on generalized formulas and average MET values without individual adjustments were found to be particularly inaccurate. The authors concluded that failure to account for individual physiological differences such as metabolic rate, body composition, and exercise efficiency undermines user confidence and reduces the credibility of digital fitness tools. These findings further reinforce the need for more personalized and adaptive estimation models in calorie tracking systems.

User trust and system transparency have also been identified as critical factors influencing the adoption and continued use of fitness monitoring technologies. Gao et al. (2015) examined user perceptions of wearable devices and app-based fitness technologies, with a particular focus on algorithmic reliability and transparency. The study found that users were more likely to trust and engage with systems that provided consistent, explainable, and scientifically grounded outputs. In contrast, inaccurate or inconsistent calorie estimates were among the primary reasons for user dissatisfaction and abandonment. Participants reported skepticism toward applications that offered results without clear explanations of how those values were generated. This highlights the importance of integrating scientific models with intelligent logic that is transparent and interpretable to users.

In addition to accuracy and transparency, usability and contextual relevance play significant roles in shaping user adoption, particularly in developing economies. Asare and Boateng (2021) examined the adoption of digital health applications among university students in Ghana and observed a growing interest in fitness and health monitoring tools. However, the study identified several challenges that limited effective usage, including interface complexity, lack of contextual relevance, and concerns about accuracy. Many participants reported that existing applications

were designed primarily for users in developed countries and did not adequately reflect local lifestyle patterns, activity types, or dietary habits. These findings suggest that successful digital health systems must consider contextual and cultural factors in addition to technical accuracy.

Similarly, Ofori (2020) emphasized the importance of developing locally relevant and transparent digital health solutions that align with user expectations and everyday experiences. The study highlighted that users are more likely to adopt and trust health applications when the systems reflect familiar activities, provide clear explanations, and offer actionable guidance. This perspective is particularly relevant for web-based solutions intended for diverse user populations, as it underscores the need for inclusivity and adaptability in system design.

Empirical evaluations of widely used commercial platforms such as MyFitnessPal, Google Fit, and Apple Health further reveal important limitations. While these platforms provide robust features for tracking physical activity and, in some cases, dietary intake, studies suggest that they often lack comprehensive integration between energy expenditure and daily caloric requirement estimation. Users are frequently presented with isolated data points, such as calories burned during exercise or calories consumed through meals, without clear guidance on how these figures relate to overall energy balance or specific fitness goals.

As a result, users are often required to interpret complex data independently, which can be challenging for individuals without prior knowledge of metabolic principles or nutritional science. This reliance on user interpretation reduces the practical usefulness of such applications and limits their effectiveness in supporting sustained behavior change. Empirical evidence indicates that users benefit most from systems that not only track data but also provide interpretive feedback and actionable recommendations tailored to individual goals.

Across the reviewed studies, several recurring themes emerge. First, accuracy in calorie estimation is consistently identified as a foundational requirement for user trust and engagement. Second, personalization and adaptability are essential for addressing individual physiological differences and lifestyle patterns. Third, transparency and explainability significantly influence user confidence in system outputs. Finally, accessibility and contextual relevance are critical for ensuring broad adoption, particularly in diverse or resource-limited settings.

Collectively, these empirical findings reveal clear gaps in existing calorie tracking and fitness monitoring systems. Despite widespread adoption, many current platforms fail to deliver accurate, personalized, and actionable insights that support effective weight management and long-term health goals. These shortcomings highlight the need for an AI-powered web application that integrates scientifically validated models with intelligent, transparent logic and user-centered design.

By synthesizing insights from empirical research, this study positions itself to address these identified gaps. The proposed system seeks to improve accuracy by incorporating validated metabolic equations, enhance personalization through adaptive logic, and foster user trust through transparency. Furthermore, its web-based architecture supports accessibility and contextual flexibility, making it suitable for diverse user populations. In doing so, the study responds directly to empirical evidence and contributes to the advancement of effective digital health and fitness monitoring solutions.

2.4 Theoretical Framework

This study is guided by the Technology Acceptance Model (TAM), a widely recognized theoretical framework for understanding and predicting user acceptance of technology. Developed by Davis (1989), TAM posits that two key factors perceived usefulness and perceived ease of use determine an individual's intention to adopt and continue using a technological system. Perceived usefulness refers to the degree to which a person believes that using a particular system will enhance their performance or help them achieve desired outcomes, while perceived ease of use refers to the extent to which a person believes that interacting with the system will be free of effort. By emphasizing these constructs, TAM provides a robust framework for evaluating factors that influence technology adoption, particularly in contexts where user engagement and sustained use are critical to achieving intended benefits.

In the context of this study, TAM serves as a guiding framework for understanding how users interact with an AI-powered web application designed to monitor energy expenditure and estimate daily caloric needs. The model helps explain how the perceived benefits of accurate, personalized, and actionable health insights influence users' willingness to adopt the system. For instance, if users perceive that the application provides reliable calorie estimates and tailored recommendations that support their fitness goals, their perceived usefulness of the system increases, thereby enhancing their intention to use it consistently. Similarly, if the application is designed to be intuitive, responsive, and easy to navigate, users' perceived ease of use will be high, further reinforcing positive behavioral intentions toward technology adoption.

TAM also provides a valuable framework for assessing the role of system design and user experience in promoting sustained engagement. Empirical studies have demonstrated that health and fitness applications often fail to retain users over time due to issues such as complexity, lack

of transparency, or insufficient feedback mechanisms (Krebs & Duncan, 2015). By applying TAM, this study emphasizes the importance of aligning system features with user expectations and capabilities. Enhancing perceived ease of use through clear interface design, step-by-step guidance, and accessible functionalities can mitigate barriers to adoption. Likewise, emphasizing the practical benefits of accurate energy expenditure monitoring and daily caloric estimation can strengthen perceived usefulness, motivating users to incorporate the application into their daily routines.

Moreover, TAM provides a foundation for integrating additional constructs that are particularly relevant in digital health contexts, such as trust, system transparency, and perceived accuracy. In health applications, users are more likely to adopt and continue using a system if they trust the underlying algorithms and understand how outputs are generated. By incorporating these considerations into the TAM framework, the study can better evaluate factors that influence user acceptance beyond the core constructs of usefulness and ease of use, thus addressing the specific challenges of AI-powered health technologies.

Finally, the use of TAM in this study supports both system development and research evaluation. During the design phase, TAM informs user-centered features that enhance usability, transparency, and perceived effectiveness. In the evaluation phase, the model provides a structured approach for assessing user attitudes, intentions, and behavioral outcomes. By grounding the study in TAM, the research ensures a theoretically informed approach to understanding technology adoption, while also offering practical guidance for designing a web application that meets user needs and promotes sustained engagement in health and fitness management.

2.4.1 Proponents of the Theory

The Technology Acceptance Model (TAM) was proposed by Fred Davis in 1989 as a theoretical framework to explain and predict how individuals come to accept and use new technologies. TAM has become one of the most widely applied models in information systems research due to its simplicity, predictive power, and ability to account for user behavior across diverse technological contexts. The model posits that the adoption of a system is primarily influenced by two key constructs: Perceived Usefulness (PU) and Perceived Ease of Use (PEOU). These constructs collectively shape a user's attitude toward the system, which in turn influences their behavioral intention to use it, ultimately determining actual system usage. By focusing on user perceptions rather than solely on technical capabilities, TAM provides a user-centered approach to understanding technology adoption.

Perceived Usefulness refers to the degree to which an individual believes that using a particular system will enhance their performance or help them achieve specific objectives. In practical terms, a system is considered useful if it delivers tangible benefits, such as increased efficiency, improved accuracy, or enhanced decision-making capabilities. For instance, in the context of digital health and fitness applications, a user is likely to perceive an AI-powered web application as useful if it provides accurate estimates of energy expenditure, personalized daily caloric needs, and actionable recommendations that support health and fitness goals. When perceived usefulness is high, users are more motivated to engage with the system, integrate it into their routines, and rely on it for decision-making.

Perceived Ease of Use, the second key construct of TAM, refers to the degree to which a user believes that interacting with a system requires minimal effort. A system that is easy to learn,

navigate, and operate reduces cognitive load and minimizes potential barriers to adoption. In the context of health applications, ease of use may involve intuitive interface design, clear data visualization, step-by-step guidance, and seamless navigation across functionalities. High perceived ease of use not only facilitates initial adoption but also contributes to sustained engagement, as users are more likely to continue using a system that is convenient and straightforward to operate.

TAM also emphasizes the interaction between these two constructs. While Perceived Ease of Use directly influences attitudes toward using the system, it can also indirectly affect Perceived Usefulness. A system that is easier to operate may be perceived as more effective because users can access and interpret its functionalities without difficulty. This interplay underscores the importance of considering both usability and practical utility when designing technological systems, particularly those intended for health and fitness applications where user motivation and adherence are critical.

Over the years, TAM has been extended and adapted to include additional factors such as perceived trust, system transparency, and social influence, particularly in contexts where user confidence and reliability are essential. In health-related applications, trust in system outputs, transparency of algorithms, and contextual relevance are crucial determinants of adoption, complementing the traditional constructs of usefulness and ease of use. By providing a structured framework to evaluate these factors, TAM allows researchers and developers to identify design elements that promote engagement, satisfaction, and long-term use.

In summary, the Technology Acceptance Model offers a robust theoretical foundation for understanding user adoption of new technologies. By highlighting the critical roles of Perceived

Usefulness and Perceived Ease of Use, TAM guides both the design and evaluation of systems, ensuring that they meet user expectations, deliver meaningful benefits, and are accessible and intuitive to operate. In the context of AI-powered health and fitness applications, TAM provides valuable insights for developing solutions that are not only technologically advanced but also widely accepted and effectively utilized by target users.

2.4.2 Strengths and Weaknesses of the Theory

A major strength of the Technology Acceptance Model (TAM) lies in its simplicity, clarity, and strong empirical validation across diverse technological domains, including information systems, mobile applications, e-learning platforms, and health information systems (Davis, 1989). The model's primary constructs Perceived Usefulness (PU) and Perceived Ease of Use (PEOU) provide a clear and intuitive framework for understanding the psychological factors that drive user acceptance of technology. By focusing on users' perceptions rather than complex technical specifications, TAM allows researchers and practitioners to identify key determinants of technology adoption without requiring extensive measurement of external variables. This simplicity makes the model highly practical, as it can be applied in both academic research and real-world system evaluation studies. Numerous empirical studies have confirmed the predictive validity of TAM, demonstrating that PU and PEOU consistently influence user attitudes, behavioral intention, and actual system use. In health and fitness contexts, TAM has been particularly valuable for understanding how users adopt mobile health applications, web-based platforms, and AI-powered monitoring tools, where individual perceptions of usefulness and ease of use strongly determine sustained engagement.

Another notable strength of TAM is its adaptability. The model can be extended or integrated with additional constructs to address specific research contexts or user populations. For example, extensions such as TAM2, TAM3, and Unified Theory of Acceptance and Use of Technology (UTAUT) incorporate factors like social influence, facilitating conditions, and user experience to provide a more nuanced understanding of technology adoption. Despite its simplicity, TAM remains sufficiently robust to serve as a foundational framework while allowing researchers to adapt it to evolving technological and contextual requirements. In applied settings, TAM provides actionable insights for system designers and developers, guiding the creation of interfaces, functionalities, and features that enhance perceived usefulness and ease of use, thereby increasing the likelihood of adoption and long-term use.

However, despite its widespread utility, TAM has several notable limitations. One major critique is that the model does not explicitly account for social, cultural, or environmental factors that may influence technology adoption. Factors such as peer influence, cultural norms, economic conditions, and access to technology can significantly affect users' willingness and ability to adopt systems, particularly in contexts such as developing countries or organizational settings. By focusing primarily on individual cognitive perceptions, TAM may overlook these external influences, limiting its explanatory power in complex or socially embedded environments.

Additionally, TAM emphasizes individual perceptions and attitudes but does not fully address organizational or contextual determinants of technology adoption. In workplace or institutional settings, factors such as management support, organizational policies, workflow integration, and infrastructure availability can play critical roles in determining system adoption. The model's omission of these factors means that while it effectively predicts user intentions at an individual

level, it may not fully capture the broader dynamics that influence actual system implementation and usage in organizational or community contexts.

Another limitation of TAM is its relative focus on initial adoption rather than post-adoption behavior or long-term engagement. While the model predicts whether users are likely to adopt a system, it provides limited guidance on understanding sustained use, satisfaction, or behavioral outcomes over time. This can be particularly relevant in health-related applications, where continued engagement is essential for achieving fitness or wellness goals. To address this gap, researchers often complement TAM with additional models or longitudinal studies that examine factors such as habit formation, trust, and perceived reliability over time.

Despite these limitations, TAM remains one of the most widely used and influential models for studying user acceptance of information systems. Its combination of simplicity, theoretical clarity, and empirical support makes it a practical and reliable framework for both research and application. In the context of AI-powered web applications for monitoring energy expenditure and daily caloric needs, TAM provides valuable guidance for understanding user perceptions, designing user-centered features, and predicting adoption. By emphasizing perceived usefulness and ease of use, the model helps ensure that the system is not only technically robust but also accessible, intuitive, and relevant to users' health and fitness objectives. Furthermore, by acknowledging TAM's limitations, researchers and developers can supplement the model with contextual or social factors, creating a more comprehensive understanding of technology adoption that supports both effective system design and sustained user engagement.

2.4.3 Application of the Theory to the Study

The Technology Acceptance Model (TAM) is highly relevant to this study because the overall success and impact of the proposed AI-powered web application are closely tied to user acceptance and engagement. While the technical accuracy and advanced functionality of the system are important, their benefits can only be realized if users are willing to adopt the application and integrate it into their daily routines. TAM provides a theoretical framework for understanding the psychological and behavioral factors that influence this adoption process, particularly through its key constructs of Perceived Usefulness (PU) and Perceived Ease of Use (PEOU). By explicitly considering these constructs in the design and evaluation of the application, the study ensures that the system is not only technologically robust but also user-centered, enhancing the likelihood of sustained engagement.

Perceived Usefulness, the first construct of TAM, is directly addressed through the application's ability to provide accurate and personalized health insights. Users of the system receive precise estimates of energy expenditure during various physical activities, as well as tailored daily caloric recommendations based on individual characteristics such as age, weight, height, and activity level. In addition, the application offers clear guidance on maintaining a healthy calorie balance, indicating whether users should aim for a caloric deficit, surplus, or maintenance depending on their fitness goals. By delivering actionable, evidence-based recommendations, the system enhances the perceived value of the application in helping users achieve their health objectives. When users perceive that the system directly supports their fitness goals and

improves decision-making regarding diet and exercise, their motivation to engage with it consistently is strengthened.

Perceived Ease of Use, the second key construct, is addressed through thoughtful design and user-centered interface features. The application emphasizes simplicity and intuitiveness, ensuring that users of varying technological literacy levels can navigate the system without difficulty. Features such as clear visualizations, step-by-step instructions, minimal technical complexity, and responsive design contribute to an effortless user experience. By minimizing cognitive and operational barriers, the system reduces user frustration and promotes seamless interaction, which is particularly important in health and fitness applications where repeated use over time is essential for achieving desired outcomes.

The integration of both Perceived Usefulness and Perceived Ease of Use creates a synergistic effect that enhances user engagement and sustained adoption. Users are more likely to continue using the system when they recognize both its practical benefits and its ease of operation. This dual focus aligns with TAM's predictive framework, which emphasizes that both constructs jointly influence users' attitudes, behavioral intentions, and actual system use. For the proposed web application, attention to these factors ensures that technical accuracy and AI-driven personalization are complemented by usability, accessibility, and transparency, all of which are critical determinants of adoption in real-world settings.

Furthermore, TAM provides a foundation for evaluating user acceptance during both the development and post-implementation phases. During system design, the model informs interface decisions, data presentation, and feedback mechanisms to optimize perceived usefulness and ease of use. In subsequent evaluation or pilot testing, TAM offers measurable

constructs for assessing user attitudes, intentions, and satisfaction, enabling developers to identify strengths and areas for improvement. By grounding system development in TAM, the study ensures that user-centered considerations are systematically integrated into both design and evaluation processes, increasing the likelihood that the application will achieve its intended impact on health and fitness behaviors.

In summary, TAM is highly relevant to this study because it directly links user perceptions and attitudes to system adoption and sustained engagement. By addressing Perceived Usefulness through accurate, personalized, and actionable recommendations, and Perceived Ease of Use through intuitive and accessible interface design, the study aims to maximize user acceptance of the AI-powered web application. This approach not only facilitates effective utilization of the system but also supports long-term behavior change in fitness and health management, ultimately enhancing the practical and societal impact of the proposed solution.

2.5 Conclusion

This chapter provided a comprehensive review of both theoretical and empirical literature relevant to the design and development of an AI-powered web application for monitoring energy expenditure and estimating daily caloric needs. The theoretical review focused on the fundamental concepts underlying energy expenditure, including the Metabolic Equivalent of Task (MET) and Basal Metabolic Rate (BMR), which serve as scientifically validated methods for calculating calories burned during physical activity and determining total daily energy requirements. Additionally, the chapter explored the role of artificial intelligence in digital health

systems, emphasizing hybrid AI approaches that combine formula-based calculations with intelligent logic to enhance personalization, transparency, and interpretability. These theoretical foundations underscore the importance of grounding technological solutions in evidence-based physiological models while leveraging AI to provide individualized recommendations that support informed decision-making and improved health outcomes.

The empirical review highlighted key insights from previous studies on fitness and calorie tracking technologies, revealing both the potential benefits and persistent challenges of existing systems. Studies by Krebs and Duncan (2015) and Li and Mensah (2023) demonstrated that while fitness applications are widely adopted, long-term user engagement is often limited due to inaccurate calorie estimations, lack of personalization, and insufficient guidance on achieving energy balance. Gao et al. (2015) emphasized that user trust is strongly influenced by algorithmic transparency and reliability, while research in developing contexts (Asare & Boateng, 2021; Ofori, 2020) highlighted the need for contextually relevant and user-friendly systems that reflect local lifestyle patterns. Collectively, these empirical findings indicate that existing platforms, including widely used applications such as MyFitnessPal, Google Fit, and Apple Health, often fail to provide integrated guidance on daily caloric requirements or actionable recommendations tailored to individual goals, limiting their effectiveness for sustainable behavior change.

In light of these insights, the chapter identified significant gaps that the proposed study seeks to address. Current systems often rely on generalized formulas that do not account for individual physiological characteristics, present outputs that lack transparency, and provide limited support for interpreting data in relation to personal fitness objectives. These limitations underscore the need for a web-based, AI-powered application that combines accurate metabolic models with

intelligent, user-centered logic, offering personalized, actionable, and easily understandable guidance for managing energy balance and supporting health and fitness goals.

To guide the development and evaluation of the proposed system, the Technology Acceptance Model (TAM) was adopted as the theoretical framework for this study. TAM emphasizes the critical roles of Perceived Usefulness and Perceived Ease of Use in shaping user attitudes, intentions, and adoption behaviors. By focusing on these constructs, the study ensures that both the practical benefits of the system such as accurate calorie estimation and tailored guidance and the usability aspects, including an intuitive interface and clear visualizations, are prioritized. This framework provides a structured approach for understanding and enhancing user acceptance, which is essential for the sustained engagement necessary to achieve meaningful health outcomes.

In summary, this chapter has established a solid academic foundation for the study by integrating insights from physiological, technological, and behavioral perspectives. The review highlighted the limitations of existing calorie tracking systems, identified opportunities for technological innovation, and justified the adoption of TAM as a guiding framework for user-centered design and evaluation. These insights form the basis for the methodological approach described in the next chapter, which outlines the research design, data collection methods, and analytical strategies that will be employed to develop and assess the proposed AI-powered web application.

CHAPTER THREE

METHODOLOGY

3.1 Introduction

This chapter presents a detailed account of the methodology adopted for the design and development of the AI-powered web application for monitoring energy expenditure and estimating daily caloric needs. The methodology provides a systematic framework for understanding how the research was conducted, how data were collected and analyzed, and how the proposed system was modeled and developed. By providing a clear and structured approach, this chapter ensures the study's rigor, replicability, and alignment with its stated objectives.

The study employed a research design that integrates both exploratory and descriptive elements, allowing for a comprehensive understanding of the requirements, functionality, and performance of the proposed application. The exploratory component facilitated the identification of gaps and limitations in existing fitness and calorie tracking systems, particularly in areas such as

personalization, transparency, and actionable guidance. This was achieved through a review of scholarly literature, analysis of popular fitness applications, and consideration of user expectations in the context of health management. The descriptive component focused on documenting the development process of the web application and evaluating its functionality in terms of usability, accuracy, and user engagement.

A mixed-methods research approach was adopted to guide the study. Qualitative methods were used to gather in-depth insights into user needs, preferences, and challenges associated with existing fitness applications. This involved collecting feedback from potential users through interviews, focus group discussions, and observations of current app usage patterns. Quantitative methods were employed to measure system performance, evaluate accuracy in calorie estimation, and assess user perceptions of usability, perceived usefulness, and ease of use. Combining qualitative and quantitative approaches allowed for a holistic understanding of both the technical and behavioral dimensions of system adoption, which is critical for the development of a user-centered AI-powered application.

The target population for this study comprised individuals who are actively engaged in physical fitness or health management, including students, working professionals, and community members interested in monitoring their diet and exercise routines. The study applied purposive sampling techniques to select participants who could provide relevant insights into user expectations, fitness behaviors, and system requirements. Data were collected using structured questionnaires, semi-structured interviews, and direct observation of user interactions with prototype interfaces. These data collection methods provided both subjective user perspectives

and objective measures of system functionality, ensuring that the application meets both technical and usability requirements.

Data analysis was conducted using a combination of statistical and thematic techniques. Quantitative data obtained from surveys and system performance tests were analyzed using descriptive statistics, such as means, percentages, and standard deviations, to summarize trends and evaluate system accuracy. Inferential statistics were also employed, where appropriate, to examine relationships between user characteristics, system usability, and perceived usefulness. Qualitative data from interviews and focus groups were analyzed thematically to identify recurring patterns, user needs, and areas for system improvement. This comprehensive approach to data analysis ensured that the application was both scientifically validated and user-centered.

In addition to research design and data collection methods, the chapter outlines the system modeling techniques employed during development. System modeling provides a visual and structured representation of how the proposed application operates, how users interact with it, and how data flows within the system. A context diagram was developed to illustrate the high-level interactions between the system and external entities, including users and third-party data sources. Use case diagrams were created to capture functional requirements and depict the various activities users can perform, such as entering personal information, tracking exercise, or receiving daily caloric recommendations. Data flow diagrams were used to represent the movement of information within the system, highlighting processes, data stores, and interaction points to ensure efficient and accurate data handling. Finally, an entity–relationship diagram was designed to model the structure of the database, detailing the relationships between key entities such as users, activities, meals, and caloric records. These modeling techniques collectively

provide a comprehensive blueprint of the system's architecture, guiding both the development and evaluation phases.

By combining rigorous research methodology with systematic system modeling, this chapter establishes a solid foundation for the design, development, and assessment of the AI-powered web application. The methodological approach ensures that the system is technically robust, user-centered, and capable of providing accurate, personalized, and actionable insights into energy expenditure and daily caloric needs. The subsequent sections of this chapter provide detailed explanations of each research and modeling component, illustrating how theoretical principles and empirical insights informed the development process, and ensuring that the proposed application aligns with both scientific and user requirements.

3.2 Research Design

The study adopts a Design and Development Research (DDR) design as the overarching research methodology, which is particularly suited for investigations that involve the conceptualization, creation, implementation, and evaluation of information systems aimed at addressing real-world problems (Peppers et al., 2007). DDR emphasizes the dual objectives of producing both practical solutions and theoretical contributions, making it an ideal approach for developing an AI-powered web application that monitors energy expenditure and calculates daily caloric needs. Unlike purely experimental or observational research designs, DDR allows for the iterative development and refinement of system features, ensuring that the resulting application is not

only technically sound but also responsive to user needs and grounded in evidence-based principles.

One of the primary advantages of DDR is its structured yet flexible approach to system development. The methodology supports iterative cycles of design, implementation, evaluation, and refinement, which are critical for addressing the complex and dynamic requirements of health and fitness applications. In the context of this study, DDR facilitates the incremental integration of scientifically validated metabolic formulas, such as Basal Metabolic Rate (BMR) and Metabolic Equivalent of Task (MET) values, with intelligent logic that adapts to individual user profiles. This iterative process allows for continuous testing and improvement of system functionalities, ensuring that calorie estimates, energy expenditure calculations, and daily caloric recommendations are accurate, personalized, and reliable. By incorporating feedback from users during development, DDR ensures that the system is aligned with practical expectations, usability standards, and behavioral goals, ultimately enhancing user satisfaction and long-term engagement.

DDR also emphasizes rigorous evaluation, which is essential for both technical validation and usability assessment. In this study, user testing is integrated at multiple stages of the development cycle to assess the functionality, accuracy, and effectiveness of the application. Test participants interact with the system to track their physical activities, enter personal data, and receive caloric recommendations, while researchers collect data on system performance, perceived usefulness, and ease of use. Quantitative analysis of accuracy metrics such as the comparison of estimated versus actual energy expenditure ensures that the system meets scientific and technical standards. Meanwhile, qualitative feedback from users informs interface

design, workflow efficiency, and the clarity of output presentations. By combining technical and user-centered evaluation, DDR provides a comprehensive framework for developing a system that is both effective and acceptable to end-users.

Furthermore, DDR supports knowledge generation by linking practical problem-solving with theoretical insights. The methodology allows researchers to explore how intelligent web-based applications can be designed to address gaps identified in existing fitness and calorie tracking technologies, including limitations in personalization, transparency, and actionable guidance. By documenting the design decisions, modeling approaches, and iterative refinements undertaken during system development, the study contributes to both applied and theoretical understanding of AI-powered health systems. This knowledge can inform future research, system improvements, and broader adoption of digital health technologies in various contexts.

In addition, DDR aligns closely with the principles of user-centered design, which are essential for applications targeting health and fitness behaviors. Health-related applications are most effective when they account for user motivations, capabilities, and constraints, in addition to delivering accurate calculations. Through the iterative cycles of DDR, the system can be refined based on user interaction patterns, feedback on usability, and evaluation of perceived usefulness and ease of use. This ensures that the AI-powered web application is not only functionally robust but also intuitive, accessible, and engaging, thereby supporting sustained use and meaningful behavior change.

In summary, the adoption of a Design and Development Research design provides a structured and comprehensive methodology for this study. DDR enables the systematic creation, testing, and refinement of the AI-powered web application, integrating scientific metabolic formulas

with intelligent logic, and ensuring that the system meets both technical standards and user expectations. The methodology supports iterative development, continuous evaluation, and evidence-based refinement, resulting in a solution that is technically accurate, user-centered, and theoretically grounded. By adopting DDR, this study effectively bridges the gap between practical problem-solving in digital health and the advancement of knowledge regarding the design, implementation, and adoption of AI-powered information systems.

3.3 Research Approach

This study employed a mixed-method research approach, integrating both quantitative and qualitative methodologies to provide a comprehensive understanding of the AI-powered web application's functionality, usability, and overall user experience. The mixed-method approach is particularly appropriate for health-related digital applications, as it allows for the assessment of both measurable system performance and the nuanced perceptions, behaviors, and preferences of users (Creswell & Plano Clark, 2018). By combining these approaches, the study ensures that the application is evaluated not only in terms of technical accuracy but also in terms of its practical relevance, usability, and alignment with user needs.

The quantitative component of the research focused on assessing the system's performance, accuracy, and usability. Structured questionnaires were administered to participants after they interacted with the application, capturing measurable data on user perceptions of ease of use, perceived usefulness, and satisfaction. Additionally, the system itself generated data during user sessions, including metrics such as calorie estimation accuracy, energy expenditure calculations,

and response times. Quantitative analysis of this data provided objective measures of system effectiveness, enabling researchers to determine whether the application met its intended technical specifications and performance standards. Statistical tools such as descriptive analysis, including mean scores, percentages, and standard deviations, were employed to summarize patterns, while inferential analyses were conducted where appropriate to examine relationships between system features and user evaluations.

In parallel, qualitative methods were applied to gain deeper insights into user interactions and experiences with the system. Observation techniques were used during testing sessions to document user behaviors, identify points of difficulty or confusion, and note interaction patterns. These qualitative observations allowed the research team to explore how users navigated the interface, interpreted outputs, and responded to recommendations regarding energy expenditure and caloric needs. In addition, informal interviews and open-ended questions provided further insight into user perceptions of the application's utility, clarity, and relevance to their personal fitness goals. By capturing these rich, contextual data, the qualitative approach complemented the quantitative findings, revealing aspects of usability and user experience that could not be fully measured through numerical indicators alone.

The integration of quantitative and qualitative approaches provides several key advantages. First, it allows for triangulation, whereby findings from different methods are compared and validated, enhancing the reliability and credibility of the results. For example, quantitative data showing high accuracy in calorie estimation can be corroborated with qualitative observations indicating that users understood and trusted the system's outputs. Conversely, discrepancies between observed user behavior and system-generated metrics can highlight areas for improvement, such

as interface design or guidance clarity. Second, the mixed-method approach enables a more holistic evaluation of the system by addressing both technical and human-centered dimensions. In digital health applications, understanding how users engage with and interpret system outputs is just as important as ensuring algorithmic precision, because sustained adoption depends on both functionality and usability.

Furthermore, the mixed-method approach supports iterative refinement of the application. Feedback from quantitative and qualitative assessments was used to identify limitations, inform design improvements, and enhance user experience. For instance, if structured questionnaires indicated that certain features were perceived as complex or confusing, observational insights could guide modifications to layout, labeling, or guidance prompts. Similarly, if system-generated data revealed discrepancies in calorie estimations for certain activities, qualitative observations could provide context for potential misunderstandings or input errors, informing adjustments to algorithms or instructional materials.

In conclusion, the adoption of a mixed-method research approach in this study ensures a comprehensive and multi-dimensional evaluation of the AI-powered web application. By combining quantitative assessments of system performance, usability, and accuracy with qualitative insights from user observations and feedback, the study captures both objective and subjective measures of effectiveness. This approach not only provides a robust basis for evaluating the technical and functional capabilities of the application but also ensures that user needs, behaviors, and experiences are prioritized. The integration of these methods is essential for the development of effective, reliable, and user-centered digital health applications, where

sustained engagement and accurate guidance are critical to achieving meaningful health and fitness outcomes.

3.4 Population

The target population for this study comprises students of the University of Professional Studies, Accra (UPSA). This population was specifically selected because students represent a demographic that is both highly engaged with digital technologies and increasingly conscious of personal health and fitness management. University students often integrate technology into multiple aspects of their daily lives, including communication, learning, entertainment, and health tracking. Their familiarity with digital tools makes them particularly suitable for interacting with AI-powered applications, as they are more likely to navigate web-based platforms with ease, provide relevant feedback on system usability, and engage meaningfully with advanced features such as personalized recommendations for caloric intake and energy expenditure.

In addition to their technological familiarity, UPSA students were chosen because they exhibit diverse fitness behaviors, lifestyle patterns, and health goals, which are essential considerations for testing an AI-powered web application. Within a university setting, students may engage in a variety of physical activities, ranging from structured exercises at campus gyms to informal sports and recreational activities. These differences in activity type, frequency, and intensity provide a rich context for evaluating the accuracy of calorie estimation algorithms, as the system must be capable of adapting to varied user profiles. Moreover, students' interest in health

management is often linked to broader lifestyle concerns such as diet, stress management, and time-constrained schedules, making them ideal participants for assessing whether the application provides actionable and practical insights into maintaining energy balance.

Another important consideration in selecting UPISA students as the target population is accessibility. University campuses provide a concentrated and readily available pool of potential participants, which facilitates efficient recruitment, observation, and testing of the system. The campus environment also allows for controlled pilot testing and user evaluation sessions, where researchers can monitor interactions, collect feedback, and make iterative improvements to the application. This accessibility is particularly advantageous for a study employing a mixed-method research approach, as it enables the collection of both quantitative data (through structured questionnaires and system-generated metrics) and qualitative data (through observation and interviews) within a manageable timeframe and resource framework.

Furthermore, focusing on students offers insights into the adoption of AI-powered health applications among young adults a demographic that is increasingly targeted by digital fitness solutions. Understanding the attitudes, perceptions, and behavioral intentions of university students provides valuable guidance for designing user-centered interfaces, tailoring personalized recommendations, and ensuring that the system is engaging and easy to use. The students' responses can inform the iterative development process, highlighting areas where the system may need enhancements in clarity, guidance, or functionality. For instance, students may provide feedback on the comprehensibility of calorie balance recommendations, the intuitiveness of navigation, or the relevance of activity tracking features, which can directly influence system refinement.

In addition, UPSA students represent a diverse population in terms of academic backgrounds, cultural perspectives, and lifestyle choices. This diversity is beneficial for evaluating the system's adaptability and generalizability. By testing the application across a range of users with different routines, dietary habits, and activity levels, the study can ensure that the system is robust and flexible enough to provide accurate and meaningful recommendations across various contexts. Insights gained from this population can also inform potential expansion of the application to other user groups, such as working professionals, community members, or students in other institutions.

In summary, the selection of UPSA students as the target population for this study is motivated by several factors. Their high engagement with digital technology, interest in fitness and health management, accessibility for pilot testing, and diversity in lifestyle behaviors make them an ideal population for evaluating the AI-powered web application. By involving this population, the study ensures that the system is tested in a realistic context, captures meaningful feedback on usability and effectiveness, and provides actionable insights for both technical refinement and user-centered design. The findings derived from this population will be instrumental in ensuring that the application is accurate, user-friendly, and capable of supporting sustained engagement in energy management and health-promoting behaviors.

3.5 Sampling and Data Collection Techniques

A purposive sampling technique was employed in this study to select participants for the usability evaluation and prototype testing of the AI-powered web application. Purposive

sampling, also referred to as judgmental or selective sampling, involves the deliberate selection of individuals who possess specific characteristics relevant to the research objectives. In the context of this study, participants were selected based on two key criteria: regular use of digital devices and active engagement in physical exercise. This approach ensures that the sample consists of individuals who are not only technologically literate but also have practical experience with fitness and health monitoring, making them capable of providing meaningful and informed feedback on system usability, functionality, and accuracy.

The rationale for using purposive sampling lies in the nature of the study and its primary objective, which is to evaluate a prototype system rather than generalize findings to a broader population. Unlike random sampling, purposive sampling allows the researcher to focus on participants who can provide rich, contextually relevant insights into system performance and user experience. In this study, participants with experience using digital fitness applications or wearable devices were prioritized, as they are more likely to understand the functionalities of similar systems and critically assess the AI-powered application's features, such as calorie estimation, daily caloric recommendations, and personalized guidance on energy balance. Their informed perspectives are essential for identifying usability issues, evaluating the clarity of outputs, and suggesting improvements to interface design and functionality.

A sample size of 20–30 participants was deemed appropriate for the study, consistent with best practices for usability evaluation and prototype testing in system development research (Gao et al., 2015). Usability studies typically require smaller, focused samples because the goal is to detect usability problems, observe user interactions, and collect qualitative and quantitative feedback rather than achieve statistical representativeness. Research in human-computer

interaction suggests that testing with 20–30 participants is sufficient to uncover the majority of usability issues, identify recurring patterns of user behavior, and provide actionable recommendations for system refinement. In addition, a sample within this range allows for efficient management of data collection, observation, and analysis while still ensuring diversity in user profiles, activity patterns, and technological familiarity.

The selection of participants also considered diversity in age, gender, academic background, and fitness levels within the target population of UPSA students. Ensuring variability in these characteristics provides a broader perspective on how different users interact with the system and perceive its usefulness and ease of use. For example, students with varying degrees of familiarity with technology may encounter different challenges in navigation, data input, or interpretation of outputs, and their feedback is valuable for designing a universally accessible and user-friendly application. Similarly, participants with different fitness routines or exercise intensities provide insights into the system's adaptability in estimating calorie expenditure across a range of activity levels.

During the recruitment process, participants were informed about the purpose of the study, the nature of the prototype, and the type of tasks they would perform during testing sessions. Ethical considerations were carefully observed, including obtaining informed consent and assuring participants of the confidentiality and security of any personal information entered into the system. By selecting participants purposefully and providing clear guidance on participation, the study ensured that data collected would be both relevant and reliable, enabling accurate evaluation of system performance and user experience.

In summary, the use of purposive sampling and the selection of a sample size of 20–30 participants provided a strategic and practical approach for evaluating the AI-powered web application. This sampling technique allowed the study to focus on individuals who were most capable of providing informed feedback on system usability, accuracy, and functionality. The chosen sample size was sufficient to identify usability issues, observe interaction patterns, and gather qualitative and quantitative insights necessary for iterative system refinement. By targeting participants with relevant technological literacy and engagement in physical activity, the study ensured that the evaluation process would yield meaningful, actionable, and user-centered findings, ultimately contributing to the development of an effective and accessible digital health application.

Data Collection Techniques

Data for this study were collected using a combination of instruments designed to capture both quantitative and qualitative information, ensuring a comprehensive evaluation of the AI-powered web application for monitoring energy expenditure and estimating daily caloric needs. The use of multiple instruments allowed for triangulation of data, enhancing the reliability, validity, and depth of the study's findings. Specifically, three main data collection instruments were employed: structured questionnaires, observation during system testing, and system-generated logs.

Firstly, structured questionnaires were administered to participants to gather essential demographic information and assess their perceptions of the system. Demographic data, including age, gender, academic program, prior experience with digital fitness applications, and

frequency of physical activity, were collected to provide context for interpreting usability and performance results. In addition to demographic information, the questionnaires included items designed to measure key constructs such as system usability, perceived usefulness, accuracy, and user satisfaction. Usability questions focused on ease of navigation, clarity of interface design, and intuitiveness of system features, while perceived usefulness items assessed the extent to which the application enhanced participants' ability to monitor calorie expenditure, track energy balance, and make informed fitness and dietary decisions. Accuracy-related questions explored participants' confidence in the system's calorie estimations and energy expenditure calculations, while satisfaction items gauged overall user experience, including whether the system met expectations and whether users would continue using it in the future. The questionnaires employed a combination of Likert-scale, multiple-choice, and open-ended items to capture both quantitative ratings and qualitative feedback, providing a balanced assessment of user perceptions.

Secondly, direct observation during system testing sessions served as a qualitative instrument to capture participants' interactions with the application in real-time. Observational techniques allowed researchers to identify usability challenges, workflow inefficiencies, and patterns in user behavior that may not have been fully captured through questionnaires alone. For example, observation provided insights into how users navigated the interface, interacted with data input forms, interpreted system outputs, and responded to recommendations regarding caloric balance. Observers also recorded instances where participants encountered difficulties, such as misinterpreting instructions, making errors in entering personal or activity data, or hesitating at certain decision points. This qualitative data was critical for understanding the practical

challenges faced by users and for identifying areas where the interface or system logic could be improved to enhance clarity, efficiency, and user satisfaction.

Thirdly, system-generated logs were employed as an objective instrument for capturing detailed records of user activity and system performance. These logs tracked interactions such as login frequency, duration of system use, navigation paths, data input patterns, and responses to prompts or recommendations. In addition, the logs recorded caloric calculations, energy expenditure estimates, and daily caloric recommendations provided by the application. System logs allowed the research team to evaluate the technical accuracy of the AI-powered calculations, verify that user inputs were correctly processed, and identify patterns in how users engaged with different features of the system. By analyzing system-generated data alongside questionnaire responses and observational notes, the study was able to triangulate findings, providing a more robust assessment of both technical performance and user experience.

The combination of these three instruments questionnaires, observation, and system logs ensured that the study captured a holistic view of the application's effectiveness. Questionnaires provided structured feedback on perceived usability, usefulness, and satisfaction; observation captured real-time interactions, behaviors, and challenges; and system logs offered objective evidence of technical accuracy, system usage patterns, and interaction frequency. Together, these instruments allowed the study to evaluate the system from multiple perspectives, ensuring that both human-centered and technical dimensions were addressed. This multi-instrument approach was particularly important for health-related digital applications, where user trust, engagement, and accuracy are all critical factors in determining the success and adoption of the system.

In summary, the carefully selected data collection instruments provided a comprehensive framework for assessing the AI-powered web application. By integrating self-reported measures, observational insights, and system-generated metrics, the study was able to capture a detailed and nuanced understanding of system performance, usability, and user experience. These instruments collectively ensured that the evaluation process was thorough, evidence-based, and capable of informing iterative refinements to the system, ultimately contributing to the development of an accurate, user-friendly, and effective digital tool for monitoring energy expenditure and managing daily caloric needs.

3.6 System Modeling and Design

To clearly represent the structure, functionality, and interactions of the proposed AI-powered web application, several system modeling techniques were employed. System modeling is a critical component of the design and development process, as it provides a visual and conceptual framework that guides both developers and stakeholders in understanding how the system operates, how data flows within it, and how users interact with its various components. By employing multiple complementary modeling techniques, the study ensured that the system architecture, processes, and data relationships were clearly defined, logically organized, and aligned with the intended functional requirements. These models also facilitate communication among developers, researchers, and end-users, enhancing transparency and ensuring that the system design addresses user needs effectively.

One of the primary modeling techniques used was the context diagram, which provides a high-level overview of the system and its interactions with external entities. The context diagram illustrates the boundaries of the system, showing how it communicates with users, databases,

third-party services, and other external systems. In the context of the AI-powered web application, the context diagram identifies key interactions, such as user input of personal and activity data, system output of caloric calculations and energy expenditure recommendations, and integration with cloud-based storage for data management. This high-level representation helps to clarify the scope of the system, highlight critical data exchanges, and ensure that all essential components and stakeholders are accounted for during the development process.

Another essential modeling technique employed was the use case diagram, which focuses on the functional requirements of the system from the perspective of end-users. The use case diagram captures the different activities that users can perform, including entering personal information, logging meals, tracking physical activities, receiving daily caloric recommendations, and viewing progress reports. Each use case is linked to the relevant actors, such as users, administrators, or the system itself, illustrating the interactions necessary to achieve specific goals. Use case diagrams are particularly useful for guiding the development of user interfaces and system workflows, as they provide a clear depiction of how users will interact with the application and the sequence of actions required to complete various tasks.

Data flow diagrams (DFDs) were also utilized to depict the movement of information within the system. DFDs provide a detailed view of how data enters, is processed, and exits the system, highlighting the relationships between different processes, data stores, and external entities. For the proposed web application, DFDs illustrate how user inputs, such as exercise logs and dietary information, are processed to calculate energy expenditure and daily caloric needs. The diagrams also show how processed data is stored in databases, retrieved for user reports, and displayed through the interface. By mapping the flow of data, DFDs help identify potential bottlenecks,

redundancies, or points of failure in the system, ensuring that information is accurately processed and securely managed.

In addition, entity–relationship diagrams (ERDs) were employed to model the structure of the system’s database and the relationships between key data entities. ERDs define entities such as users, activities, meals, and caloric records, and specify how these entities are related to one another. For example, an ERD may show that each user can have multiple activity logs or meal entries, while each activity log is associated with specific MET values for accurate energy expenditure calculations. This modeling technique is essential for designing a robust and normalized database structure, ensuring that data is consistently organized, easily retrievable, and accurately linked to the corresponding system functions. ERDs also support system scalability, allowing for the seamless addition of new features or entities without compromising database integrity.

By employing these modeling techniques context diagrams, use case diagrams, data flow diagrams, and entity–relationship diagrams the study was able to comprehensively represent the proposed system’s architecture, functionality, and data relationships. Each technique addresses a different aspect of system design, from high-level interactions and functional requirements to detailed data processing and database structure. Together, these models provide a clear, cohesive blueprint that guides system development, supports iterative testing and refinement, and ensures that the application meets both technical and user-centered objectives. Furthermore, these visual representations facilitate communication among the development team, stakeholders, and end-users, enhancing understanding, collaboration, and transparency throughout the development lifecycle. Ultimately, the use of multiple modeling techniques ensures that the AI-powered web

application is well-structured, logically organized, and capable of delivering accurate, personalized, and actionable insights into energy expenditure and daily caloric needs.

3.6.1 Context Diagram

The context diagram provides a high-level view of the system and its interaction with external entities.

Description:

The AI-powered calorie monitoring system interacts primarily with users, who input personal details and workout information. The system processes this data and returns caloric estimates and recommendations.

External Entities:

- User
- Firebase Database

System: AI-Powered Calorie Monitoring Web Application

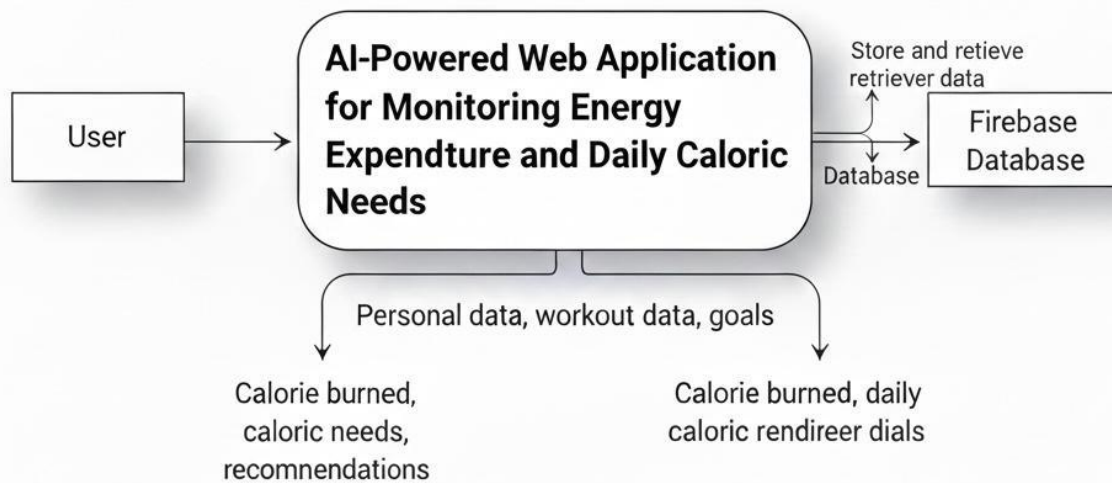
Inputs:

- User profile data

- Workout details

Outputs:

- Calories burned
- Daily caloric needs
- Fitness recommendations



3.6.2 Use Case Diagram

The use case diagram illustrates how users interact with the system.

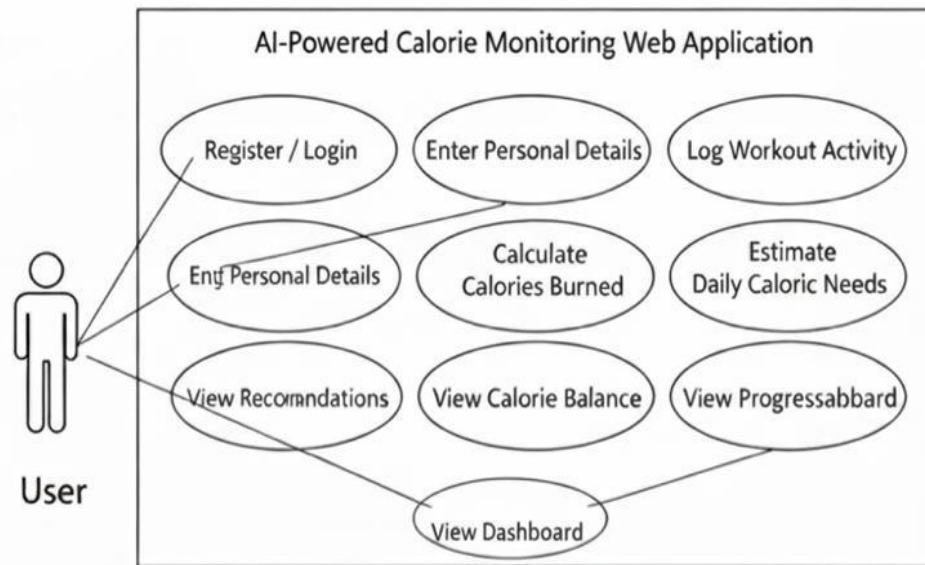
Actors:

- User

Use Cases:

- Register/Login
- Enter personal details
- Log workout activity
- Calculate calories burned
- Estimate daily caloric needs
- View calorie balance (deficit/surplus)
- View progress and recommendations

This diagram emphasizes user-centered system functionality and aligns with the Technology Acceptance Model by focusing on ease of use and usefulness (Davis, 1989).



3.6.3 Data Flow Diagram (DFD)

The Data Flow Diagram shows how data moves through the system.

Processes include:

1. User Authentication

2. Activity Logging

3. Calorie Calculation

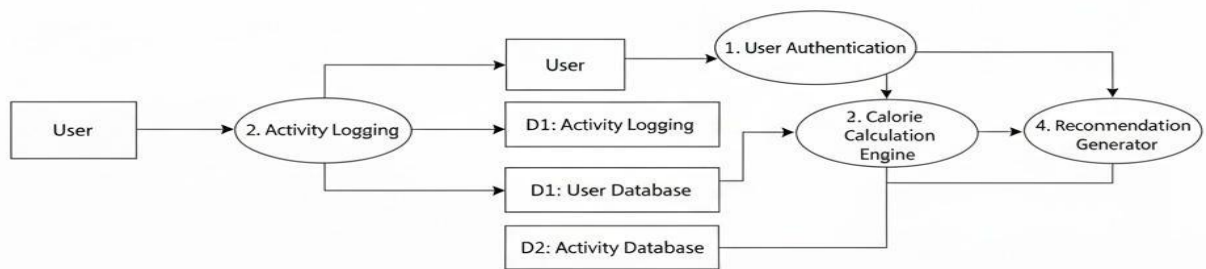
4. Recommendation Generation

Data Stores:

- User Database
- Activity Records
- Calorie History

Data Flow:

User inputs data → System processes data using formulas and AI logic → Results stored in database → Feedback displayed to user.



3.6.4 Entity–Relationship (ER) Diagram

The ER diagram represents the database structure of the system.

Entities include:

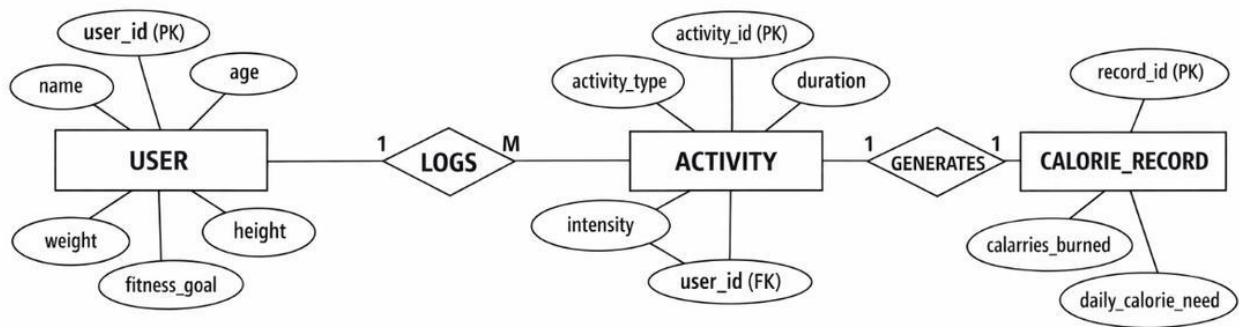
- User
- Activity
- CalorieRecord

- Recommendation

Relationships:

- A user performs many activities
- Each activity generates calorie records
- Each calorie record produces recommendations

This structure supports efficient data storage, retrieval, and analysis.



3.7 Method of Data Analysis

The analysis of data collected in this study was conducted using a combination of quantitative and qualitative methods, consistent with the mixed-method research approach adopted for evaluating the AI-powered web application for monitoring energy expenditure and estimating daily caloric needs. This dual approach enabled a comprehensive assessment of both measurable system performance and user experience, providing insights into technical functionality, usability, and the overall acceptance of the application by participants.

Quantitative Data Analysis

Quantitative data were primarily derived from structured questionnaires administered to participants following their interaction with the system. These questionnaires captured information on several key constructs, including system usability, perceived usefulness, accuracy of caloric calculations, and overall user satisfaction. Descriptive statistical methods were employed to analyze the quantitative data, including the calculation of frequencies, percentages, mean values, and standard deviations. These statistical measures provided a clear summary of participants' responses, allowing the researchers to identify prevailing trends, assess the general perception of the system, and evaluate the extent to which the application met user expectations.

Frequencies and percentages were used to summarize categorical responses, such as participants' agreement or disagreement with statements related to usability and satisfaction. For instance, data on ease of navigation, clarity of interface design, and intuitiveness of system features were categorized and expressed as percentages to indicate the proportion of participants who rated

each aspect positively, neutrally, or negatively. This approach allowed the research team to identify specific features that were well-received, as well as those requiring improvement.

Mean values and standard deviations were computed for Likert-scale items to provide an average measure of user perceptions and the degree of variability among participants. For example, mean scores for perceived usefulness offered insight into whether participants felt the system effectively supported their fitness and dietary decision-making. Standard deviations helped gauge the consistency of user responses, highlighting areas where opinions were more divergent and potentially indicating differences in user expectations or experiences. These descriptive statistics not only quantified the overall performance of the system but also provided a foundation for identifying patterns that could inform iterative design improvements.

Qualitative Data Analysis

Qualitative data were primarily obtained through direct observation of participants during system testing sessions. Observations focused on capturing user interactions with the application, including navigation patterns, response to prompts, data input behaviors, and any difficulties encountered during use. Unlike questionnaires, which rely on self-reported perceptions, observational data provided real-time insights into how participants actually engaged with the system, allowing researchers to identify subtle usability challenges and interaction issues that may not be fully captured through structured responses.

Thematic analysis was employed to analyze the qualitative data, following a systematic process of coding, categorizing, and identifying recurring patterns or themes. Initial coding involved assigning descriptive labels to observed behaviors, such as “difficulty entering activity data,”

“confusion with calorie output interpretation,” or “smooth navigation of interface.” These codes were then grouped into broader categories reflecting key usability aspects, such as “data input challenges,” “interface clarity,” “user engagement,” and “interpretation of recommendations.” Finally, overarching themes were derived from these categories, highlighting common patterns and critical insights related to system performance and user experience.

The thematic analysis provided valuable context for interpreting the quantitative findings. For instance, if a questionnaire indicated moderate satisfaction with system usability, observation data could reveal the underlying reasons, such as confusing labels, unclear instructions, or navigation inefficiencies. Similarly, themes emerging from qualitative analysis could highlight differences in interaction patterns based on participants’ prior experience with digital fitness applications or varying familiarity with health-related terminology. By integrating qualitative insights with quantitative results, the study achieved a richer and more nuanced understanding of system performance and user acceptance.

Integrated Analysis and Interpretation

The combination of quantitative and qualitative analyses enabled triangulation of findings, enhancing the credibility and comprehensiveness of the evaluation. Quantitative measures provided objective evidence of the system’s perceived usability, usefulness, and satisfaction levels, while qualitative observations offered explanatory insights into user behavior, challenges, and interaction patterns. For example, if a high percentage of participants reported that the system was useful in tracking caloric intake, observational data could illustrate how users interacted with specific features, such as meal logging or activity entry, to achieve that usefulness. Conversely, discrepancies between quantitative satisfaction scores and observed

interaction difficulties could inform targeted improvements to enhance the overall user experience.

Furthermore, the integrated analysis allowed for the identification of both strengths and weaknesses in the application. Strengths, such as accurate caloric calculations, intuitive interface elements, and meaningful recommendations for energy balance, were confirmed through consistent quantitative and qualitative feedback. Weaknesses, such as points of confusion during data entry, navigation inefficiencies, or lack of clarity in interpreting outputs, were highlighted through observed behaviors and thematic coding. This comprehensive evaluation not only validated the technical accuracy of the system but also provided actionable insights for refining usability, enhancing interface design, and improving user guidance.

In summary, the analysis of data collected in this study utilized a rigorous and multi-faceted approach. Quantitative data were analyzed using descriptive statistics, including frequencies, percentages, and mean values, to evaluate system usability, perceived usefulness, and user satisfaction. Qualitative data obtained through observations were subjected to thematic analysis to identify recurring patterns related to user behavior, interaction flow, and usability challenges. The combined analysis provided both objective and subjective perspectives, ensuring a thorough evaluation of system performance, accuracy, and acceptance. By integrating these analytical approaches, the study was able to generate meaningful insights that informed iterative system refinement, supported user-centered design, and contributed to the development of a functional, accurate, and user-friendly AI-powered web application for monitoring energy expenditure and managing daily caloric needs.

3.8 Conclusion

This chapter presented a detailed overview of the methodological framework employed for the design, development, and evaluation of the AI-powered web application for monitoring energy expenditure and determining daily caloric needs. The methodology was carefully structured to ensure that the system was developed in a systematic, rigorous, and user-centered manner, with the dual objectives of achieving technical accuracy and promoting high user engagement and satisfaction. By combining well-established research designs, a mixed-method approach, purposive sampling, structured data collection instruments, system modeling techniques, and robust data analysis procedures, the study established a comprehensive and reliable foundation for both the development and evaluation of the application.

The chapter began by outlining the research design, which adopted a Design and Development Research (DDR) approach. DDR is particularly suited for studies involving the creation, implementation, and evaluation of information systems aimed at solving real-world problems. In this study, DDR facilitated the iterative design, testing, and refinement of the web application, ensuring that both functional correctness and user needs were addressed. The design approach allowed the research team to integrate scientifically validated metabolic equations with AI-powered logic, enabling personalized calorie tracking and energy expenditure monitoring. By employing DDR, the study maintained a strong focus on system usability, accuracy, and alignment with the intended objectives of supporting informed health and fitness decisions.

A mixed-method research approach was adopted to capture both quantitative and qualitative insights into system performance and user experience. Quantitative methods, primarily through structured questionnaires, provided measurable data on usability, perceived usefulness, accuracy,

and overall user satisfaction. These metrics allowed the study to quantify the system's effectiveness and provide evidence of its technical reliability. Qualitative methods, particularly direct observation of participants during system testing, captured real-time interactions, usability challenges, and patterns in user behavior. The integration of these methods facilitated triangulation of data, enhancing the reliability, validity, and comprehensiveness of the evaluation process. This mixed-method approach was particularly appropriate for health-related digital applications, where both objective performance and subjective user experience are critical determinants of system adoption and success.

The target population for the study consisted of students from the University of Professional Studies, Accra (UPSA). This population was chosen due to their frequent use of digital technologies, interest in fitness and health management, and accessibility for pilot testing. The study focused on individuals who regularly engage in physical activity and have prior experience with digital devices, ensuring that participants were capable of interacting meaningfully with the AI-powered web application. By selecting participants with diverse activity routines, academic backgrounds, and technological literacy, the study was able to evaluate the system's adaptability and generalizability, while also gathering feedback that could inform user-centered refinements.

A purposive sampling technique was employed to select participants who met specific inclusion criteria, including regular use of digital devices and active engagement in physical exercise. A sample size of 20–30 participants was considered sufficient for usability testing and prototype evaluation, consistent with established recommendations for human-computer interaction studies. Purposive sampling allowed the researchers to focus on participants who could provide informed, relevant, and actionable feedback on system usability, functionality, and accuracy. The sampling

strategy also ensured diversity in user experiences and perspectives, which was critical for identifying potential usability challenges and evaluating the overall effectiveness of the system.

Data collection instruments included structured questionnaires, observational protocols, and system-generated logs. Questionnaires captured demographic information and quantitative measures of usability, perceived usefulness, accuracy, and satisfaction. Observations provided qualitative insights into user behavior, navigation patterns, interaction challenges, and real-time system usage, highlighting areas for improvement that might not be evident from self-reported data alone. System logs offered objective data on user interactions, including login frequency, duration of use, activity logging, and the accuracy of calorie calculations. The triangulation of these instruments ensured a thorough evaluation of both technical performance and user experience, enabling the identification of strengths, weaknesses, and areas for iterative refinement.

To provide a clear and structured representation of the system, system modeling techniques were employed, including context diagrams, use case diagrams, data flow diagrams (DFDs), and entity-relationship diagrams (ERDs). Context diagrams illustrated the high-level interactions between the system and external entities, while use case diagrams detailed user activities and functional requirements. DFDs mapped the flow of information within the system, highlighting how data was processed, stored, and retrieved, and ERDs defined the database structure and relationships between key data entities. These models provided a visual and conceptual framework for both developers and stakeholders, ensuring clarity in system design, data management, and user interactions.

Finally, data analysis methods were detailed to ensure that findings could be interpreted systematically and rigorously. Quantitative data from questionnaires were analyzed using descriptive statistics, including frequencies, percentages, and mean values, to summarize usability, perceived usefulness, and user satisfaction. Qualitative data from observations were analyzed using thematic analysis, identifying recurring patterns and themes related to user behavior, interaction flow, and usability challenges. By combining quantitative and qualitative analyses, the study achieved a holistic understanding of system performance, highlighting areas of strength and identifying opportunities for enhancement.

In conclusion, this chapter provided a comprehensive overview of the methodological framework employed in the study. The research design, approach, population, sampling, data collection instruments, system modeling techniques, and data analysis methods collectively ensured that the AI-powered web application was developed in a systematic, evidence-based, and user-centered manner. By integrating these methodological components, the study established a strong foundation for the subsequent stages of the project, which involve system implementation, evaluation, and iterative refinement based on user feedback and performance metrics. The methodology described in this chapter provides confidence that the resulting application will be both technically robust and highly usable, capable of accurately monitoring energy expenditure and supporting personalized daily caloric management for users.

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